

DRILLING MACHINE

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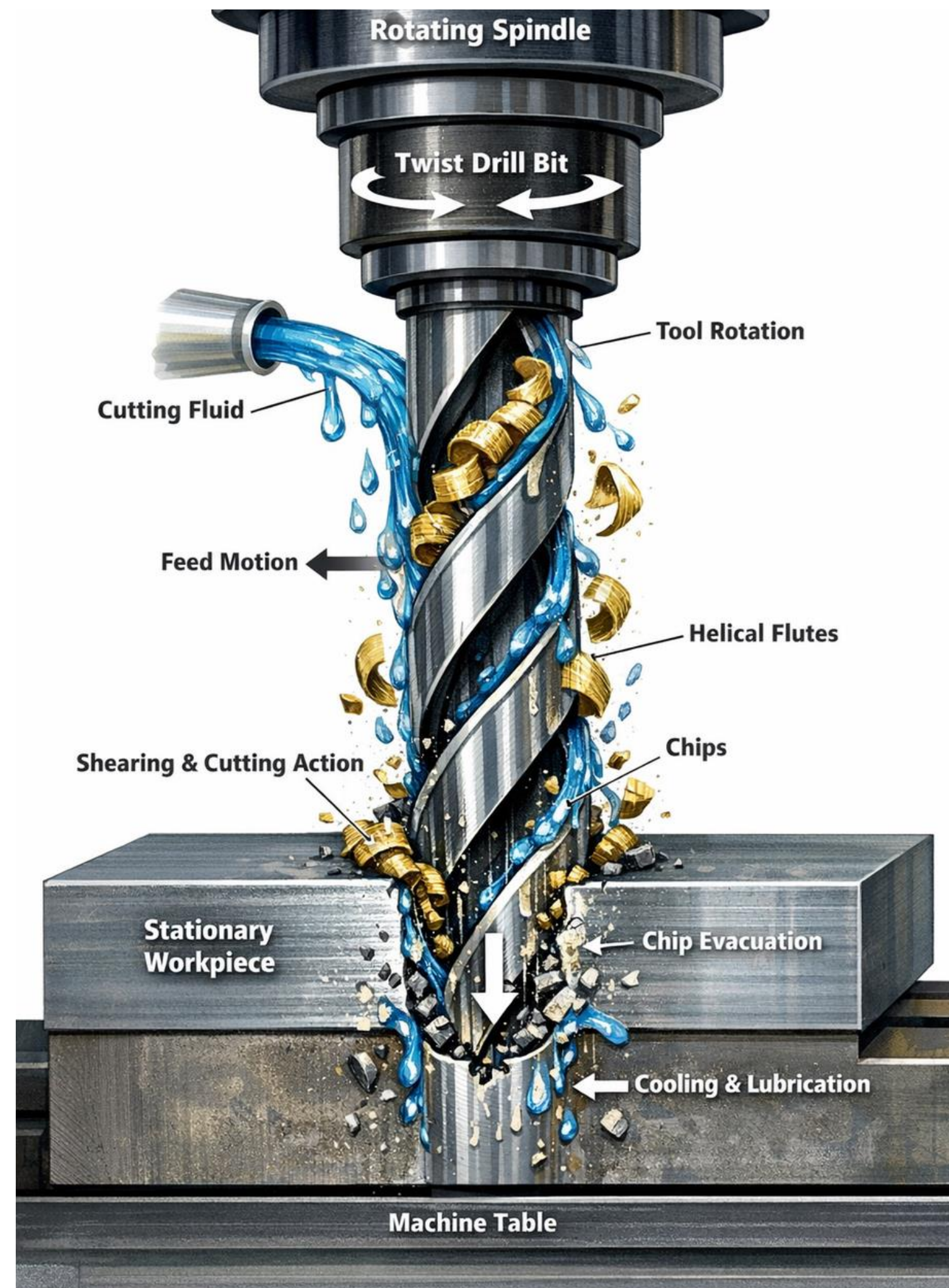
- **BRIEF INTRODUCTION**
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INTRODUCTION

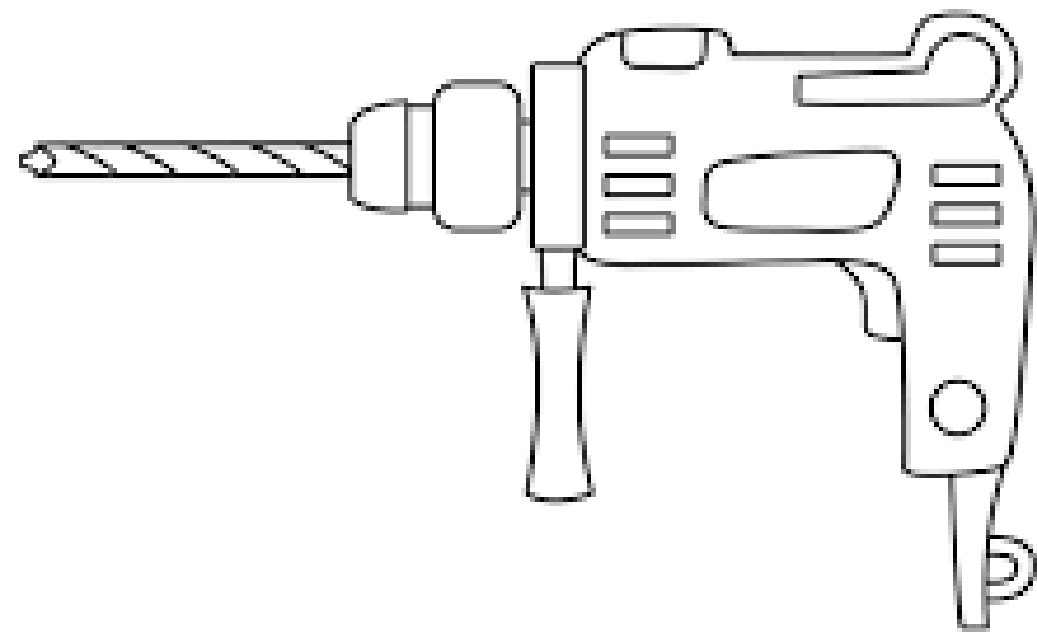
- A robust, stationary power tool essential in manufacturing and fabrication.
- Designed specifically to hold and rotate a cutting tool (drill bit) at high speeds.
- Features a mechanism to apply controlled vertical, linear force (feed) to the workpiece.
- Offers significantly higher precision, stability, and power compared to handheld drilling tools.
- Capable of handling a variety of materials, from soft woods to hardened steel, depending on the tool and machine specifications.

BASIC WORKING PRINCIPLE

- **STATINARY WORKPIECE:** The material is rigidly clamped to the work table and does not move.
- **TOOL ROTATION:** The machine spindle rotates the cutting tool (drill bit) at a high speed.
- **LINEAR FEED:** The rotating tool is fed axially downward into the workpiece.
- **SHEARING ACTION:** The drill's cutting edges compress and shear the material to generate the hole.
- **CHIP EVACUATION:** The helical flutes on the drill bit continuously lift chips out of the hole.
- **COOLING:** Flutes also provide a pathway for cutting fluids to reach the cutting zone, preventing overheating.

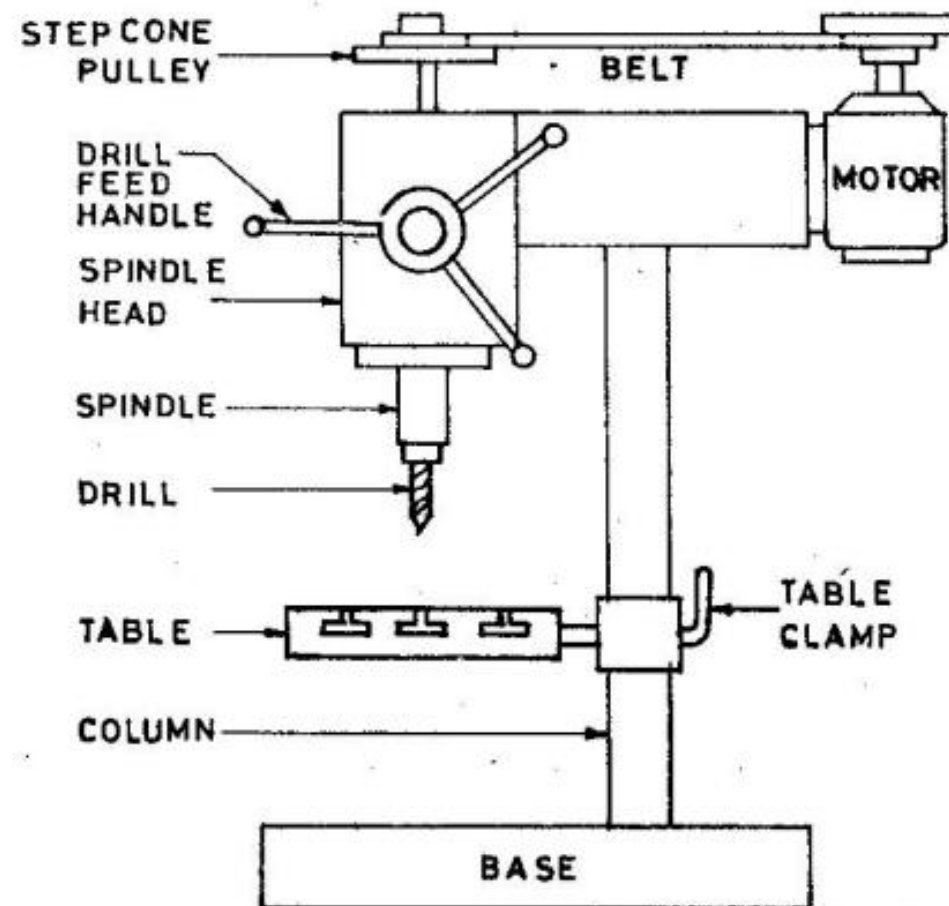


TYPES OF DRILLING MACHINE



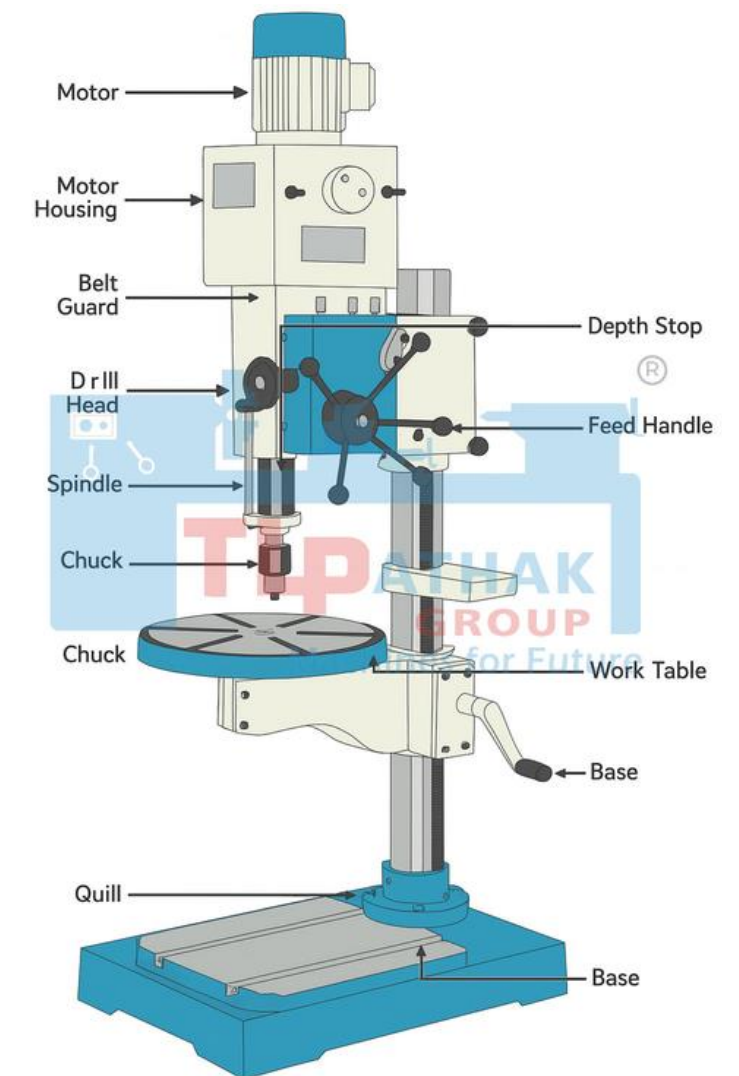
1. PORTABLE DRILLING MACHINE

Light-weight, hand-held machine used for small drilling tasks at different locations. Common in repair and maintenance work.



2. SENSITIVE DRILLING MACHINE

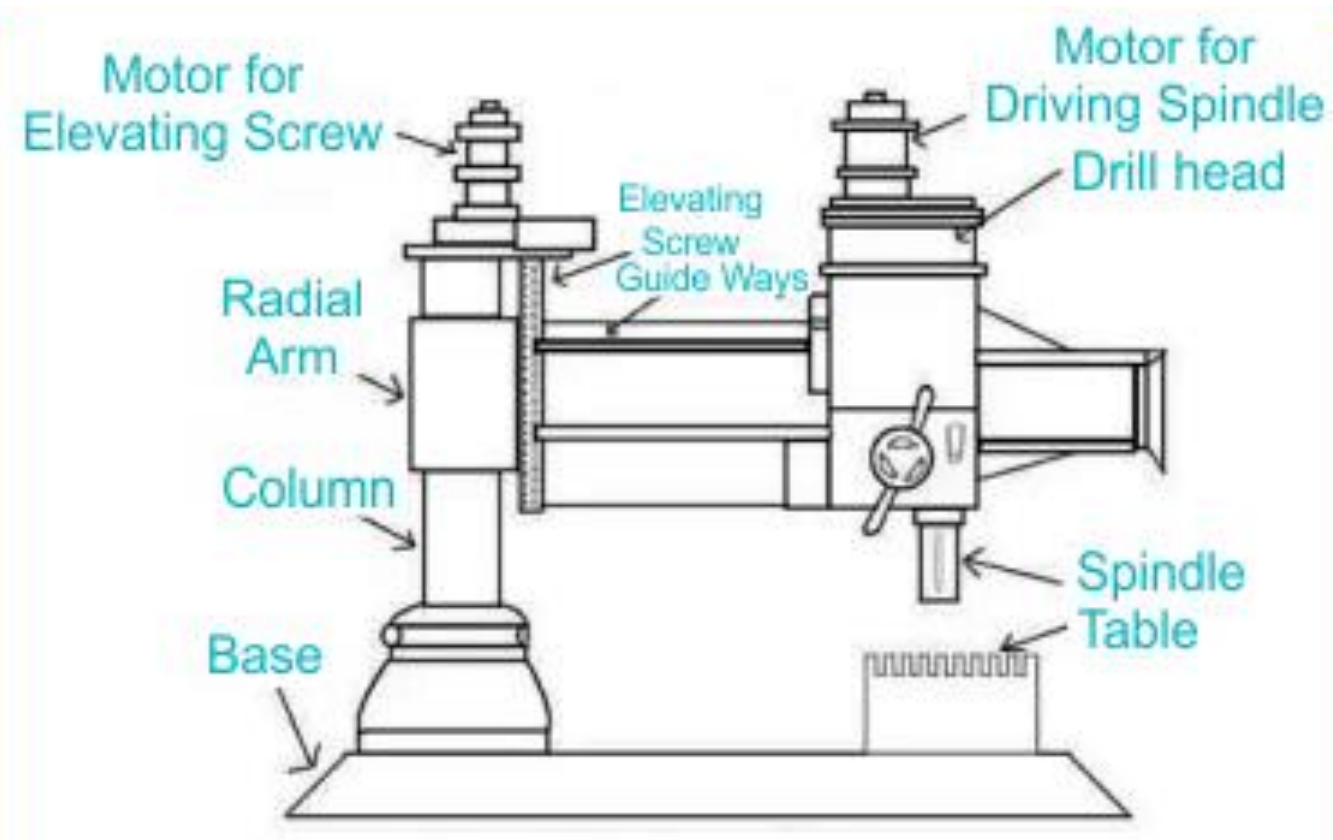
Bench-mounted machine used for precise, light-duty work. Operator controls the feed manually, allowing better accuracy for small holes (up to 15.5 mm).



3. PILLAR DRILLING MACHINE

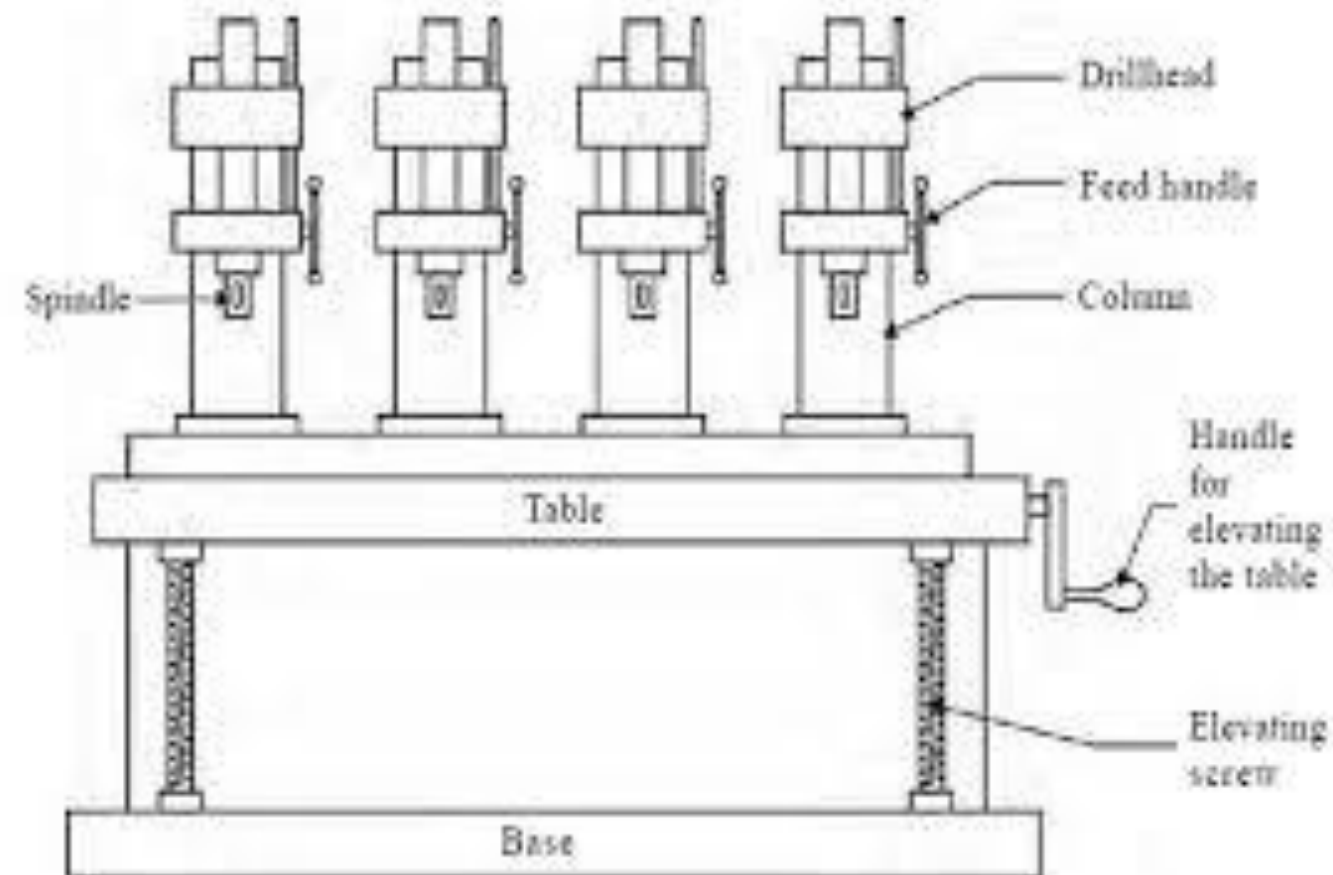
Heavy, floor-mounted machine suitable for medium-sized jobs. It can handle larger drill bits and usually has an automatic (power) feed mechanism.

SOME ADVANCED DRILLING MACHINE



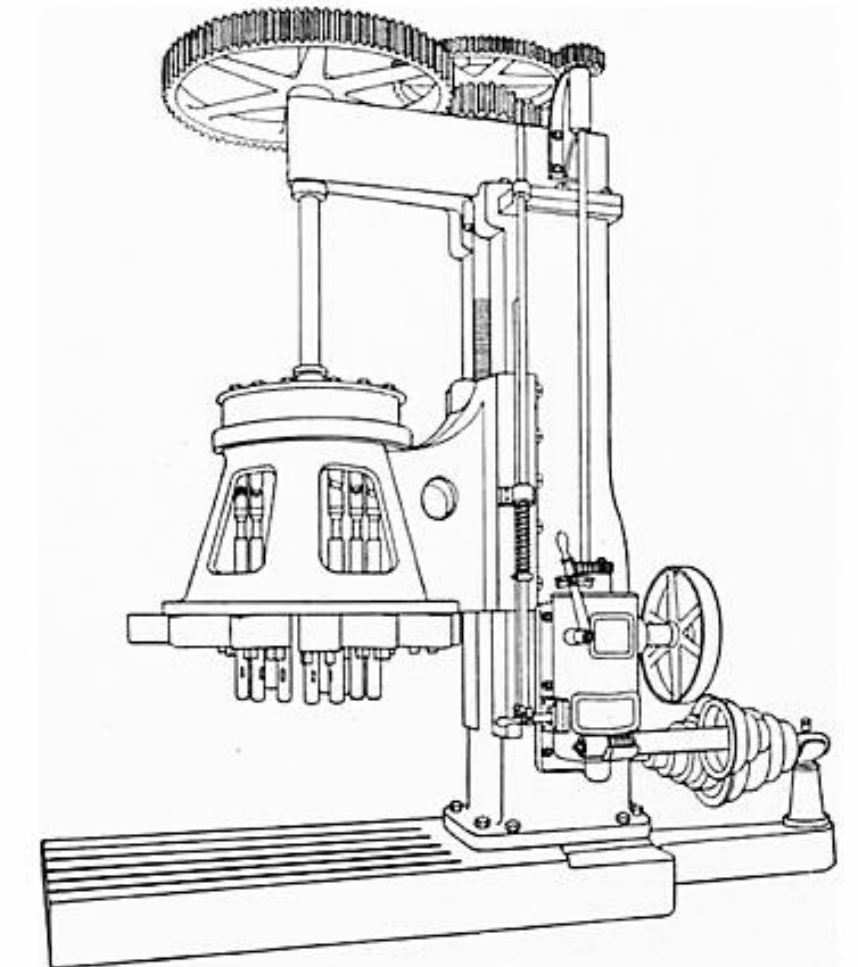
1. RADIAL DRILLING MACHINE

Used for large and heavy workpieces. The drill head can move along a radial arm, which rotates around a column, making it very flexible.



1. GANG DRILLING MACHINE

Has multiple drilling heads arranged in a row on a single table. Each spindle performs different operations one after another (drilling, reaming, tapping).

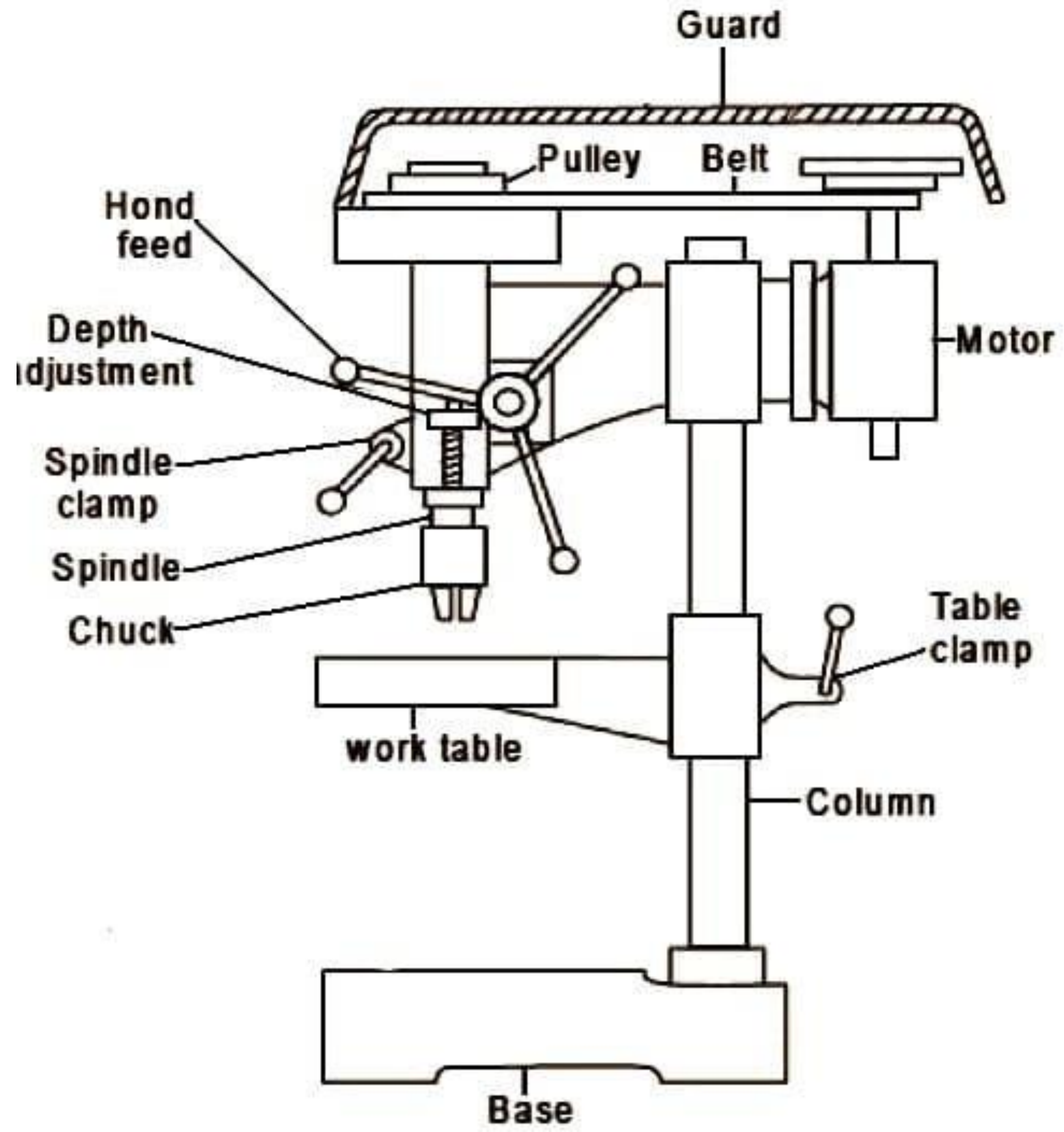


1. MULTIPLE SPINDLE DRILLING MACHINE

Contains several spindles that operate simultaneously, allowing multiple holes to be drilled in one operation. Ideal for mass production.

CONSTRUCTION

- **BASE & COLUMN:** Heavy cast-iron foundation for stability; the vertical pillar supports all assemblies.
- **WORK TABLE:** Adjustable, rigid platform used to securely clamp the workpiece.
- **DRIVE MECHANISM:** Motor and pulleys (or gearbox) used to control the spindle speed (RPM).
- **SPINDLE & QUILL:** The spindle provides rotation; it sits inside the non-rotating quill, which provides linear up/down feed.
- **DRILL CHUCK:** The clamping device that securely holds the drill bit.
- **FEED HANDLE:** Manual lever used to plunge the rotating tool into the material.



ADVANTAGES

- High-speed hole generation compared to other methods.
- Versatile: can perform operations like reaming, tapping, and counterboring.
- Available in a wide range of sizes, from portable to heavy-duty radial.
- Relatively simple to operate and maintain.
- Highly accurate when paired with jigs and fixtures.

DISADVANTAGES

- Primarily limited to creating cylindrical holes.
- Hole surface finish is relatively rough; often requires secondary operations (like reaming) for precision.
- Long, deep holes can cause drill wandering or deflection.
- Chip removal can be difficult in deep holes, requiring frequent "peck" drilling

APPLICATIONS

- Creating rivet and bolt holes in structural steel.
- Machining engine blocks and mechanical casings.
- Preparing holes for internal threading (tapping).
- Creating countersinks for flush screw seating.
- Widely used in fabrication, automotive, and aerospace industries.

CUTTING PARAMETERS

- **CUTTING SPEED (V):** The peripheral speed of the drill at its outside diameter. Measured in m/min.
- **FEED RATE (f):** The distance the drill advances into the workpiece per revolution. Measured in mm/rev.
- **DEPTH OF CUT (d):** Half of the drill diameter for a standard drilling operation. $d = D/2$ (where D is drill diameter)

IMPORTANT FORMULAE

- SPINDLE SPEED (N): -

$$N = \frac{(1000 * V)}{(\pi * D)}$$

- MACHINING TIME (T):-

$$T = \frac{L}{f * N}$$

- TOTAL LENGTH OF TOOL TRAVEL (L):-

$$L = \text{Hole Depth} + \text{Approach Length} + \text{Over-travel}$$

- MATERIAL REMOVAL RATE (MRR):-

$$MRR = \frac{(\pi * D^2) * f * N}{4}$$

Q. A HOLE OF 20 MM DIAMETER IS TO BE DRILLED THROUGH A STEEL PLATE OF 50 MM THICKNESS. THE RECOMMENDED CUTTING SPEED IS 25 M/MIN AND THE FEED RATE IS 0.2 MM/REV. ASSUME AN APPROACH LENGTH OF 3 MM AND AN OVER-TRAVEL OF 2 MM.FIND :-

A)The required spindle speed in RPM.

B)The total machining time.

Ans:- Given: D = 20 mm, Thickness = 50 mm, V = 25 m/min, f = 0.2 mm/rev.

- Total Length (L): Thickness + Approach + Over-travel
$$L = 50 + 3 + 2 = 55 \text{ mm.}$$

- Spindle Speed (N):

$$N = \frac{(1000 \times 25)}{(\pi \times 20)}$$

or, N = 397.88 RPM (or approx 398 RPM)

- Machining Time (T):

$$T = \frac{55}{(0.2 \times 398)}$$

or, T = 0.69 minutes (or approx 41.4 seconds)

UNSOLVED EXAMPLES

1. A through-hole of 25 mm diameter is to be drilled in a low-carbon steel plate of 40 mm thickness. The selected cutting speed is 30 m/min and the feed rate is 0.25 mm/rev. Assuming an approach length of 3 mm and an over-travel of 4 mm, determine:

- a) The required rotational speed of the spindle (in RPM).
- b) The total time required to drill the hole.
- c) The Material Removal Rate (MRR).

2. A blind hole of 12 mm diameter and 35 mm depth is to be drilled in a cast iron block. The drill has a point angle of 118° . The spindle speed is set to 600 RPM and the feed is 0.15 mm/rev. Assuming an approach distance of 2 mm, calculate:

- a) The total length of tool travel (Hint: you will need to account for the drill point length, which is approximately 0.3 times D for a 118° drill point).
- b) The total machining time in seconds.

The background features a light blue and white color scheme with abstract geometric elements. Teal lines form various shapes, including rectangles and triangles, some of which are nested or overlapping. In the corners, there are clusters of small teal circles arranged in a grid pattern. The central text 'THANK YOU' is rendered in a bold, black, distressed font.

THANK YOU